Xiao Tan

B.S. Student in Computer Science, University of North Carolina at Chapel Hill
Undergraduate Research Assistant, Hardware Security Lab @ UNC
tanxiao@unc.edu • xtan0611@gmail.com • Google Scholar • Website • LinkedIn

RESEARCH INTERESTS

Hardware Security • Formal Methods • Computer Architecture

EDUCATION

University of North Carolina at Chapel Hill, Chapel Hill, NC, United States — August 2023 — May 2027
Bachelor of Science in Computer Science — Cumulative GPA: 3.94/4.00
Dean's List (2023 Fall - 2026 Spring)

PUBLICATIONS

CHARGE: Leveraging CWE Hierarchies for Hardware Security SystemVerilog Assertion Generation.
Xiao Tan, Cynthia Sturton.
Accepted to the IEEE/ACM International Conference on Computer-Aided Design (ICCAD), 2026.

Hardware Security Benchmarks for Open-Source SystemVerilog Designs.
Jayden Rogers, Niyaz Shakeel, **Xiao Tan**, Samantha Espinosa, Divya Mankani, Cade Chabra, Kaki Ryan, Cynthia Sturton.
In *Proceedings of the IEEE Secure Development Conference (SecDev)*, 2025.

RESEARCH EXPERIENCE

Hardware Security Lab @ UNC — Chapel Hill, NC, United States
PI: Professor Cynthia Sturton — Fall 2024 – Present

CHARGE (ICCAD 2026) — Fall 2025 – Present

- Designed and implemented CHARGE, an automated framework that generates hardware security SystemVerilog Assertions (SVAs) directly from RTL using Common Weakness Enumeration (CWE) hierarchies and large language models, eliminating the need for trusted specifications.
- Designed a three-stage pipeline comprising asset identification, behavior mining, and SVA generation. Leveraged the hierarchical structure of the CWE taxonomy to improve security-critical asset identification.
- Evaluated CHARGE on the Hack@DAC18, 19, and 21 benchmark suites, detecting 27 of 42 known hardware security bugs and identifying one previously unknown bug.
- Currently extending CHARGE to reduce false positives and improve the quality of generated security assertions.

Hardware Security Benchmarks (SecDev 2025) — Spring 2025

- Developed SystemVerilog Assertions (SVAs) to verify security properties of open-source System-on-Chip (SoC) designs using Cadence JasperGold Formal Property Verification (FPV).
- Assessed the coverage of manually written security properties across hardware security weakness classes using the hierarchical structure of the Common Weakness Enumeration (CWE) taxonomy.

Research Infrastructure — Fall 2025

- Collaborated on maintaining SylQ-SV, a symbolic execution engine for SystemVerilog, by resolving parser compatibility issues between PyVerilog and PySlang.

ACADEMIC PROJECTS

Verification of Access Control Enforcement in AXI Interconnect — UNC Chapel Hill
COMP 637: Formal Methods for System Security — Spring 2026

- Verified access-control enforcement in an AXI-Lite interconnect extracted from the Hack@DAC21 OpenPiton SoC using assertion-based verification.
- Translated 21 access-control property templates from the AKER framework (Restuccia et al., ICCAD 2021).
- Applied Cadence JasperGold FPV and SPV engines to perform model checking and information-flow tracking across controller–peripheral interactions.
- Successfully instantiated 17 properties, with 12 satisfied and covered; analyzed 5 violations and identified two inaccuracies in the original AKER templates.

FPGA Rhythm Game on Custom MIPS Processor UNC Chapel Hill
COMP 541: Digital Logic and Computer Design Spring 2026

- Designed and implemented a custom single-cycle MIPS processor in SystemVerilog and deployed it on a Nexys A7 FPGA. Integrated VGA graphics, accelerometer, keyboard, and audio output through memory-mapped I/O (MMIO).
- Developed a real-time rhythm game inspired by Trombone Champ, featuring VGA-rendered scrolling notes, accelerometer-based pitch control, keyboard-triggered note timing, score/combo tracking, and discrete audio feedback.
- Implemented the game in C and manually translated the program into MIPS assembly for execution on the custom processor.
- Built the complete hardware/software stack, including peripheral drivers, graphics rendering, input handling, and audio generation, demonstrating hardware/software co-design on an FPGA platform.

Network Traffic Application Classification via Sequence Models UNC Chapel Hill
COMP 431: Internet Services and Protocols Fall 2025

- Collected real-world network traffic traces using `tcpdump` and constructed a dataset across multiple application types (web, video, gaming, VoIP)
- Extracted structural traffic features including packet sizes, inter-arrival times (IAT), and MTU-based metrics
- Evaluated clustering methods (K-Means, DBSCAN) and observed limited separation across application classes
- Applied sequence models (BiLSTM/LSTM) to capture temporal patterns in traffic flows, achieving ~96% classification accuracy

PRESENTATIONS

Poster: Leveraging CWE Hierarchies for Hardware Security SystemVerilog Assertion Generation
CRA UR2PhD Undergraduate Mentoring Workshop and Research Showcase April 2026
Selected participant (travel fully funded).

SELECTED COURSES

Formal Verification for System Security; Digital Logic and Computer Design; Computer Security Concepts; Models of Languages and Computation; Operating Systems (Fall 2026); Compilers (Fall 2026)

AWARDS

Summer Undergraduate Research Fellowship (SURF) UNC Chapel Hill
Awarded a $4,000 summer research stipend to work on automated hardware security assertion generation. March 2026

TEACHING EXPERIENCE

UNC-Chapel Hill Department of Computer Science Chapel Hill, NC, United States
Undergraduate Teaching Assistant June 2025 — December 2025

- Supported 300+ students in COMP 301 (Object-Oriented Programming and Design Patterns) through 10+ weekly office hours across Summer and Fall 2025
- Assisted students in debugging complex Java codebases involving OOP, MVC architecture, and concurrency
- Designed programming assignments and exam review resources as part of the Additional Materials Committee

OTHER EXPERIENCE

UNC-Chapel Hill ITS Service Desk Chapel Hill, NC, United States
Student Assistant - Walk-in Services June 2025 — December 2025

- Diagnosed and resolved 50+ monthly technical support tickets across hardware, software, and network issues
- Achieved a high first-contact resolution rate by delivering clear, step-by-step technical solutions
- Documented recurring issues to streamline troubleshooting workflows and improve support efficiency

SKILLS

- **Hardware Description Languages:** Verilog, SystemVerilog
- **Formal Verification:** SystemVerilog Assertions (SVA), Cadence JasperGold FPV/SPV
- **Programming Languages:** Python, Java, C/C++
- **Tools:** Vivado, Git, Linux